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For Ubuntu 24.04 LTS on VMware Workstation . . . 3

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ScrapOS ERP - Complete Installation Guide

For Ubuntu 24.04 LTS on VMware Workstation

Document Version: 1.1 (Updated)
Date: February 2026
Target System: Ubuntu 24.04.3 LTS on VMware Workstation 17
Access: Local LAN + VPN

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---

1. INTRODUCTION

This guide will help you install ScrapOS ERP (Scrap Metal Trading System) on your local Ubuntu server. The guide is written for non-IT persons with step-by-step instructions.

What you will have after installation: - A fully working ERP system for scrap metal trading - Accessible from any computer on your local network - Accessible via VPN when outside the office - Automatic daily backups

Time Required: Approximately 2-3 hours for complete setup

---

2. SYSTEM REQUIREMENTS

Minimum Hardware (for VMware Virtual Machine):

Component	Minimum	Recommended
CPU Cores	2 cores	4 cores
RAM	4 GB	8 GB
Disk Space	40 GB	100 GB
Network	Bridged Adapter	Bridged Adapter

Host Computer Requirements:

- Windows 10/11 with VMware Workstation 17 installed
- At least 16 GB RAM (to share with VM)
- At least 100 GB free disk space

---

3. VMWARE VIRTUAL MACHINE SETUP

Step 3.1: Download Ubuntu Server ISO

1. Open your web browser
2. Go to: https://ubuntu.com/download/server
3. Click "Download Ubuntu Server 24.04.3 LTS"
4. Save the file (approximately 2 GB)

Step 3.2: Create New Virtual Machine

1. Open VMware Workstation 17
2. Click "File" > "New Virtual Machine"
3. Select "Typical (recommended)" > Click "Next"
4. Select "Installer disc image file (iso)"
5. Click "Browse" and select the Ubuntu ISO you downloaded
6. Click "Next"

Step 3.3: Configure Virtual Machine

1. **Full Name:** ScrapOS Server
2. **User name:** scrapos
3. **Password:** Choose a strong password (WRITE IT DOWN!)
4. Click "Next"

Step 3.4: Name and Location

1. **Virtual machine name:** ScrapOS-ERP
2. **Location:** Leave default or choose your preferred folder
3. Click "Next"

Step 3.5: Disk Configuration

1. **Maximum disk size:** 100 GB
2. Select "Store virtual disk as a single file"
3. Click "Next"

Step 3.6: Customize Hardware (IMPORTANT)

1. Click "Customize Hardware"
2. **Memory:** Set to 8192 MB (8 GB)
3. **Processors:** Set to 4
4. **Network Adapter:** Select "Bridged" (this allows LAN access)
5. Click "Close"
6. Click "Finish"

Step 3.7: Start Installation

1. Click "Power on this virtual machine"
2. Wait for Ubuntu installation to complete (15-30 minutes)
3. When prompted, press ENTER to reboot
4. Login with username "scrapos" and your password

---

4. UBUNTU SERVER INSTALLATION

The VMware easy install will handle most of the Ubuntu installation automatically.

If you see a manual installation screen, follow these steps:

1. Select "English" as language
2. Select "Install Ubuntu Server"
3. Select your keyboard layout
4. For network, select "Use DHCP" (automatic IP)
5. Leave proxy blank
6. Use default mirror
7. Select "Use entire disk"
8. Confirm the installation

9. Create user account:
    - Your name: ScrapOS Admin
    - Server name: scrapos-server
    - Username: scrapos
    - Password: (choose strong password)
10. Skip Ubuntu Pro
11. Install OpenSSH server: YES (select with spacebar, then Enter)
12. Skip featured snaps
13. Wait for installation to complete
14. Select "Reboot Now"

---

5. INITIAL SERVER CONFIGURATION

After the server reboots, login with your username and password.

Step 5.1: Update the System

Type these commands one by one, pressing Enter after each:

```
sudo apt update
```

(Enter your password when asked)

```
sudo apt upgrade -y
```

(Wait for this to complete - may take 5-10 minutes)

Step 5.2: Find Your Server's IP Address

```
ip addr show
```

Look for a line like: `inet 192.168.1.XXX` **WRITE DOWN THIS IP ADDRESS** - you will need it later!

Example: 192.168.1.100

Step 5.3: Set a Static IP Address (Recommended)

This ensures your server always has the same IP address.

```
sudo nano /etc/netplan/50-cloud-init.yaml
```

Delete everything and paste this (change IP addresses to match your network):

```yaml
network:
  version: 2
  ethernets:
    ens33:
      dhcp4: no
      addresses:
        - 192.168.1.100/24
```

```
    routes:
      - to: default
        via: 192.168.1.1
    nameservers:
      addresses:
        - 8.8.8.8
        - 8.8.4.4
```

IMPORTANT: - Change `192.168.1.100` to your desired IP - Change `192.168.1.1` to your router's IP (usually ends in .1) - `ens33` might be different - check with `ip addr show`

Save the file: - Press `Ctrl + X` - Press `Y` - Press `Enter`

Apply the changes:

```
sudo netplan apply
```

---

6. INSTALLING REQUIRED SOFTWARE

Step 6.1: Install Basic Tools

```
sudo apt install -y curl wget git build-essential
```

Step 6.2: Install Node.js 20 (for Frontend)

```
curl -fsSL https://deb.nodesource.com/setup_20.x | sudo -E bash -
sudo apt install -y nodejs
```

Verify installation:

```
node --version
```

Should show: v20.x.x

```
npm --version
```

Should show: 10.x.x

Step 6.3: Install Yarn (Package Manager)

```
sudo npm install -g yarn
```

Verify:

```
yarn --version
```

Step 6.4: Install Python 3.11 (for Backend)

```
sudo apt install -y python3.11 python3.11-venv python3-pip
```

Verify:

```
python3.11 --version
```

Should show: Python 3.11.x

Step 6.5: Install PM2 (Process Manager)

```
sudo npm install -g pm2
```

This will keep your application running even after restart.

---

7. INSTALLING MONGODB DATABASE

IMPORTANT: MongoDB 8.0 for Ubuntu 24.04 (Noble)

Step 7.1: Import MongoDB 8.0 GPG Key

```
curl -fsSL https://www.mongodb.org/static/pgp/server-8.0.asc | sudo gpg -o /usr/share/keyrings/
```

Step 7.2: Add MongoDB 8.0 Repository for Ubuntu 24.04

```
echo "deb [ arch=amd64,arm64 signed-by=/usr/share/keyrings/mongodb-server-8.0.gpg ] https://rep
```

Step 7.3: Update Package List

```
sudo apt update
```

Step 7.4: Install MongoDB

```
sudo apt install -y mongodb-org
```

Step 7.5: Start MongoDB Service

```
sudo systemctl start mongod
sudo systemctl enable mongod
```

Step 7.6: Verify MongoDB is Running

```
sudo systemctl status mongod
```

You should see "active (running)" in green.

Press `q` to exit the status screen.

Step 7.7: Test MongoDB Connection

```
mongosh
```

You should see the MongoDB shell. Type `exit` to close it.

Troubleshooting MongoDB Installation

If you get "unable to locate package mongodb-org" error:

Fix 1: Remove any old repository files

```
sudo rm /etc/apt/sources.list.d/mongodb-org-7.0.list 2>/dev/null
sudo rm /etc/apt/sources.list.d/mongodb*.list 2>/dev/null
```

Fix 2: Re-run Steps 7.1 to 7.4

Fix 3: Clean apt cache

```
sudo apt clean
sudo apt update
```

---

8. SETTING UP SCRAPOS ERP

Step 8.1: Create Application Directory

```
sudo mkdir -p /opt/scrapos
sudo chown scrapos:scrapos /opt/scrapos
cd /opt/scrapos
```

Step 8.2: Get the Application Files

Option A: If you have the ZIP file:

First, install unzip:

```
sudo apt install -y unzip
```

Copy the ZIP file to server using SCP or WinSCP, then:

```
unzip scrapos-erp.zip -d /opt/scrapos
```

Option B: If you saved to GitHub:

```
git clone https://github.com/YOUR_USERNAME/YOUR_REPO.git /opt/scrapos
```

Step 8.3: Verify File Structure

```
ls -la /opt/scrapos
```

You should see: - `backend/` folder - `frontend/` folder - Other files

---

9. CONFIGURING THE APPLICATION

Step 9.1: Configure Backend

```
cd /opt/scrapos/backend
```

Create environment file:

```
nano .env
```

Paste the following content:

```
MONGO_URL=mongodb://localhost:27017
DB_NAME=scrapos_erp
JWT_SECRET=your-super-secret-key-change-this-to-something-random-at-least-32-characters
JWT_ALGORITHM=HS256
ACCESS_TOKEN_EXPIRE_MINUTES=1440
```

IMPORTANT: Change the JWT_SECRET to a random string of letters and numbers (at least 32 characters). Example: `mY5up3rS3cr3tK3y2024Scr4p0sERP!@#`

Save and exit: `Ctrl+X`, then `Y`, then `Enter`

Step 9.2: Install Backend Dependencies

```
cd /opt/scrapos/backend
python3.11 -m venv venv
source venv/bin/activate
pip install --upgrade pip
pip install -r requirements.txt
```

Step 9.3: Configure Frontend

```
cd /opt/scrapos/frontend
```

Create environment file:

```
nano .env
```

Paste the following (replace IP with your server's IP):

```
REACT_APP_BACKEND_URL=http://192.168.1.100:8001
```

Save and exit: `Ctrl+X`, then `Y`, then `Enter`

Step 9.4: Install Frontend Dependencies

```
cd /opt/scrapos/frontend
yarn install
```

(This may take 5-10 minutes)

Step 9.5: Build Frontend for Production

```
yarn build
```

(This may take 2-5 minutes)

---

10. STARTING THE SERVICES

Step 10.1: Install Serve (for Frontend)

```
cd /opt/scrapos/frontend
yarn add serve
```

Step 10.2: Create PM2 Configuration

```
cd /opt/scrapos
nano ecosystem.config.js
```

Paste this content:

```
module.exports = {
  apps: [
    {
      name: 'scrapos-backend',
      cwd: '/opt/scrapos/backend',
      script: 'venv/bin/uvicorn',
      args: 'server:app --host 0.0.0.0 --port 8001',
      interpreter: 'none',
      env: {
        PATH: '/opt/scrapos/backend/venv/bin:' + process.env.PATH
      }
    },
    {
      name: 'scrapos-frontend',
      cwd: '/opt/scrapos/frontend',
      script: 'node_modules/.bin/serve',
      args: '-s build -l 3000',
      interpreter: 'none'
    }
  ]
};
```

Save and exit: `Ctrl+X`, then `Y`, then `Enter`

Step 10.3: Start All Services

```
cd /opt/scrapos
pm2 start ecosystem.config.js
```

Step 10.4: Verify Services are Running

```
pm2 status
```

You should see both services showing "online" in green.

Step 10.5: View Logs (if needed)

```
pm2 logs
```

Press `Ctrl+C` to stop viewing logs.

---

11. SETTING UP AUTO-START ON BOOT

Step 11.1: Save PM2 Configuration

```
pm2 save
```

Step 11.2: Generate Startup Script

```
pm2 startup
```

This will display a command. **COPY AND RUN** the exact command shown.

Example (yours may be different):

```
sudo env PATH=$PATH:/usr/bin pm2 startup systemd -u scrapos --hp /home/scrapos
```

Step 11.3: Test Auto-Start

Reboot the server:

```
sudo reboot
```

After reboot, login and check:

```
pm2 status
```

Both services should be running.

---

12. ACCESSING FROM OTHER COMPUTERS

Step 12.1: Test Local Access

On the server itself:

```
curl http://localhost:3000
curl http://localhost:8001/api/health
```

Step 12.2: Access from Another Computer on LAN

On any computer connected to the same network: 1. Open a web browser (Chrome, Firefox, Edge) 2. Go to: `http://192.168.1.100:3000` (Replace with your server's IP address)

Step 12.3: First Time Setup

1. Click "Register" to create your admin account
2. Fill in:
    - Full Name: Your Name
    - Email: admin@yourcompany.com

- Password: (choose strong password)
- Role: Admin

3. Click "Create Account"
4. You will be logged in automatically

Step 12.4: VPN Access

If you have a VPN configured: 1. Connect to your VPN 2. Access using the server's internal IP: `http://192.168.1.100:3000`

---

13. FIREWALL CONFIGURATION

Step 13.1: Enable UFW Firewall

```
sudo ufw enable
```

(Type `y` when asked)

Step 13.2: Allow Required Ports

```
sudo ufw allow 22/tcp
sudo ufw allow 3000/tcp
sudo ufw allow 8001/tcp
```

Step 13.3: Verify Firewall Rules

```
sudo ufw status
```

You should see:

```
22/tcp     ALLOW    Anywhere
3000/tcp   ALLOW    Anywhere
8001/tcp   ALLOW    Anywhere
```

---

14. BACKUP PROCEDURES

Step 14.1: Create Backup Directory

```
sudo mkdir -p /opt/scrapos/backups
sudo chown scrapos:scrapos /opt/scrapos/backups
```

Step 14.2: Create Backup Script

```
nano /opt/scrapos/backup.sh
```

Paste this content:

```bash
#!/bin/bash

# Configuration
BACKUP_DIR="/opt/scrapos/backups"
DATE=$(date +%Y%m%d_%H%M%S)
DB_NAME="scrapos_erp"

# Create backup directory
mkdir -p $BACKUP_DIR

# Backup MongoDB
mongodump --db $DB_NAME --out $BACKUP_DIR/db_$DATE

# Compress backup
cd $BACKUP_DIR
tar -czf backup_$DATE.tar.gz db_$DATE
rm -rf db_$DATE

# Keep only last 30 days of backups
find $BACKUP_DIR -name "backup_*.tar.gz" -mtime +30 -delete

echo "Backup completed: backup_$DATE.tar.gz"
```

Save and exit: `Ctrl+X`, then `Y`, then `Enter`

Step 14.3: Make Script Executable

```bash
chmod +x /opt/scrapos/backup.sh
```

Step 14.4: Test Backup

```bash
/opt/scrapos/backup.sh
```

Step 14.5: Schedule Daily Automatic Backup

```bash
crontab -e
```

If asked, choose nano (option 1).

Add this line at the end:

```
0 2 * * * /opt/scrapos/backup.sh >> /opt/scrapos/backups/backup.log 2>&1
```

This runs backup every day at 2:00 AM.

Save and exit: `Ctrl+X`, then `Y`, then `Enter`

Step 14.6: Restore from Backup

If you need to restore:

```
cd /opt/scrapos/backups
tar -xzf backup_YYYYMMDD_HHMMSS.tar.gz
mongorestore --db scrapos_erp db_YYYYMMDD_HHMMSS/scrapos_erp
```

---

15. TROUBLESHOOTING GUIDE

Problem: Cannot Access from Other Computers

Check 1: Verify services are running

```
pm2 status
```

Check 2: Verify firewall allows connections

```
sudo ufw status
```

Check 3: Verify IP address

```
ip addr show
```

Check 4: Test from server

```
curl http://localhost:3000
curl http://localhost:8001/api/health
```

Problem: Services Won't Start

Check logs:

```
pm2 logs scrapos-backend
pm2 logs scrapos-frontend
```

Restart services:

```
pm2 restart all
```

Problem: MongoDB Not Running

Check status:

```
sudo systemctl status mongod
```

Restart MongoDB:

```
sudo systemctl restart mongod
```

Check MongoDB logs:

```
sudo tail -100 /var/log/mongodb/mongod.log
```

Problem: "Connection Refused" Error

1. Check if backend is running on correct port:

```
sudo netstat -tlnp | grep 8001
```

2. Check if frontend is running:

```
sudo netstat -tlnp | grep 3000
```

Problem: Slow Performance

1. Check system resources:

```
htop
```

(Press `q` to exit. Install with: `sudo apt install htop`)

2. Check disk space:

```
df -h
```

3. Increase VM memory in VMware settings

Problem: "unable to locate package mongodb-org"

This happens when wrong repository is configured.

Solution:

```
# Remove old repository
sudo rm /etc/apt/sources.list.d/mongodb*.list

# Add correct repository for Ubuntu 24.04
curl -fsSL https://www.mongodb.org/static/pgp/server-8.0.asc | sudo gpg -o /usr/share/keyrings/

echo "deb [ arch=amd64,arm64 signed-by=/usr/share/keyrings/mongodb-server-8.0.gpg ] https://rep

sudo apt update
sudo apt install -y mongodb-org
```

---

16. QUICK REFERENCE COMMANDS

Service Management

Action	Command
Start all services	`pm2 start all`
Stop all services	`pm2 stop all`
Restart all services	`pm2 restart all`
View status	`pm2 status`
View logs	`pm2 logs`
View specific log	`pm2 logs scrapos-backend`

System Commands

Action	Command
Update system	`sudo apt update && sudo apt upgrade -y`
Reboot server	`sudo reboot`
Shutdown server	`sudo shutdown now`
Check disk space	`df -h`
Check memory	`free -h`
Check IP address	`ip addr show`

MongoDB Commands

Action	Command
Start MongoDB	`sudo systemctl start mongod`
Stop MongoDB	`sudo systemctl stop mongod`
Restart MongoDB	`sudo systemctl restart mongod`
Check status	`sudo systemctl status mongod`
Open MongoDB shell	`mongosh`

Backup Commands

Action	Command
Manual backup	`/opt/scrapos/backup.sh`
List backups	`ls -la /opt/scrapos/backups/`
View backup log	`cat /opt/scrapos/backups/backup.log`

APPENDIX A: SERVER INFORMATION (Fill In)

Item	Value
Server IP Address	_____
Server Username	scrapos
Server Password	_____
MongoDB Port	27017
Frontend URL	http://[IP]:3000
Backend API URL	http://[IP]:8001

Admin Account

Item	Value
Admin Email	_____
Admin Password	_____

APPENDIX B: NETWORK DIAGRAM

```
YOUR OFFICE NETWORK
===================

   Router (192.168.1.1)
          |
          |
   -------|-------
   |             |
   |             |
VMware Host   Other PCs
   |          (Access via browser)
   |          http://192.168.1.100:3000
   |
Ubuntu VM (ScrapOS)
IP: 192.168.1.100
   |
   |-- Frontend :3000
   |-- Backend  :8001
   |-- MongoDB  :27017


VPN Users: Connect to VPN, then access http://192.168.1.100:3000
```

APPENDIX C: CHECKLIST

After installation, verify:

- ☐ Ubuntu 24.04 installed on VMware
- ☐ Static IP configured
- ☐ Node.js installed (v20.x)
- ☐ Python 3.11 installed
- ☐ MongoDB 8.0 installed and running
- ☐ Backend running on port 8001
- ☐ Frontend running on port 3000
- ☐ Firewall configured
- ☐ Auto-start on boot configured
- ☐ Backup schedule configured
- ☐ Can access from other computers on LAN
- ☐ Admin account created

END OF INSTALLATION GUIDE

Document Version: 1.1 Updated: February 2026 MongoDB Version: 8.0 for Ubuntu 24.04 (Noble)